my New Dataset:

Daily (High/Max) Temperature (℃)
9.3
5.6
7.8
7.2
8.1
7.5
7.0
8.4
8.2
7.4

9.3+5.6+7.8+7.2+8.1+7.5+7.0+8.4+8.2+7.4=76.5
76.5/10=7.65℃=mean(average)

Mode=n/a or none (most frequent)

Median=7.65 (middle number when arranged from smallest to largest)

2. **Calculate the squared differences from the mean for each data point:**

(9.3-7.65)=1.65*1.65=2.7225℃

(5.6-7.65)=-2.05*-2.05=4.2025

(7.8-7.65)=0.15*0.15=0.0225

(7.2-7.65)=-0.45*-0.45=0.2025

(8.1-7.65)=0.45*0.45=0.2025

(7.5-7.65)=-0.15*-0.15=0.0225

(7.0-7.65)=-0.65*-0.65=0.4225

(8.4-7.65)=0.75*0.75=0.5625

(8.2-7.65)=0.55*0.55=0.3025

(7.4-7.65)=-0.25*-0.25=0.0625

3. **Calculate the average of these squared differences (variance):**

2.7225+4.2025+0.0225+0.2025+0.2025+0.0225+0.4225+0.5625+0.3025+0.0625=8.725 (sum of the squared differences)

8.725/10=0.8725℃=variance

4. **Take the square root of the variance to get the standard deviation:**

Sqroot of 0.8725=0.93

The standard deviation of the daily (high/max) temperature is **0.93**℃. This means that on average, the daily (high/max) temperature deviates from the mean by **0.93**℃.

If we replace the following 2 data points in the dataset:

Old Temperature → **New Temperature**
9.3℃ → **4.8**℃
5.6 → **4.5**

4.8+**4.5**+7.8+7.2+8.1+7.5+7.0+8.4+8.2+7.4=70.9
70.9/10=7.09℃ = mean(average)

Mode = n/a or none (most frequent)

Median = 7.45 (middle number when arranged from smallest to largest)

3. **Calculate the squared differences from the mean for each data point:**

(4.8-7.09) = -2.29*-2.29=5.2441℃

(4.5-7.09) = -2.59*-2.59=6.7081

(7.8-7.09) =0.71*0.71=0.5041

(7.2-7.09) = 0.11*0.11=0.0121

(8.1-7.09) = 1.01*1.01=1.0201

(7.5-7.09) = 0.41*0.41=0.1681

(7.0-7.09) = -0.09*-0.09=0.0081

(8.4-7.09) = 1.31*1.31=1.7161

(8.2-7.09) = 1.11*1.11=1.2321

(7.4-7.09) = 0.31*0.31=0.0961

4. **Calculate the average of these squared differences (variance):**

5.2441+6.7081+0.5041+0.0121+1.0201+0.1681+0.0081+1.1761+1.2321+0.0961=16.709 (sum of the squared differences)

16.709/10 =1.6709℃ = variance

5. **Take the square root of the variance to get the standard deviation:**

Sqroot of 1.6709 = 1.29

The standard deviation of the daily (high/max) temperature is **1.29**. This means that on average, the daily (high/max) temperature deviates from the mean by **1.29**℃.

Comparison and Analysis

	my New Dataset	**my New Dataset (with 2 data points replaced)**
Mean	**7.65**℃	**7.09**℃
Mode	**n/a or none**	**n/a or none**
Median	**7.65**	**7.45**
Variance	**0.8725**	**1.6709**
Standard Deviation	**0.93**	**1.29**

Measures of Central Tendency

The **mean** daily (high/max) temperature of the **first dataset** is **greater** than that of the second dataset, because two of the data points in the first dataset are greater than those in the second dataset, with the rest being the same.

There is **no mode** daily (high/max) temperature **in either dataset**, because no data point occurs most frequently/often (more than once).

The **median** daily (high/max) temperature of the **first dataset** is **greater** than that of the second dataset, because the 5th and 6th data points (ordered) in the first dataset are 7.5℃ and 7.8℃ and the 5th and 6th data points (ordered) in the second dataset are 7.4℃ and 7.5℃.

Measures of Dispersion

The **variance** in the daily (high/max) temperature of the **first dataset** is **smaller** than that of the second dataset, because most of the differences between data points are smaller than in the second dataset (most of them start with 7℃ or 8℃). This means the data points are generally closer to each other in the first dataset than in the second dataset.

Because the variance of the first dataset is smaller than that of the second dataset, the **standard deviation** of the daily (high/max) temperature in the **first dataset** is also **lower** than that of the second dataset. This means the data points are generally closer to, and more centered around, the mean in the first dataset than in the second dataset.

In other words, the data points 4.8℃ and 4.5℃ are less “typical” in/of the first dataset than the data points 8.4℃ and 5.4℃ are in/of the second dataset.